

IAS2026 Bulletproof Closure

Final executive report | branch research/ias2026-bulletproof-closure-v1

Disposition: the frozen IEEE-39 nominal closure survives, the stronger historical V9 transfer expectation is refuted, and robustness, second-model, same-mechanism PowerDynamics, and new blind-holdout claims are explicitly untested or stopped by scope.

This report distinguishes document-specified expectations from measured results. No expected value was retuned after observation, and the user's pre-existing dirty checkout was preserved in a separate linked worktree.

1. Results that survived

- Gate 0 reproduces the frozen P4 H4 nominal alpha: $0.1270064666836382 \text{ s}^{-1}$ versus 0.1270065 s^{-1} .
- The frozen P4 structure reports 15 stable proper subsets.
- Four randomized theorem/property-test modules pass, including the port bridge and transverse construction.
- The frozen boundary and nonlinear audit reproduce retrospectively: $g^*=0.2076814044$, $f=0.7064247879 \text{ Hz}$, and 32/32 declared TDS agreements.
- The documented governor changes P4 H4 alpha to $-0.07454098428813066 \text{ s}^{-1}$.
- PowerDynamics 5.0.0 reaches its official IEEE-39 tutorial equilibrium: 39 buses and 46 branches.

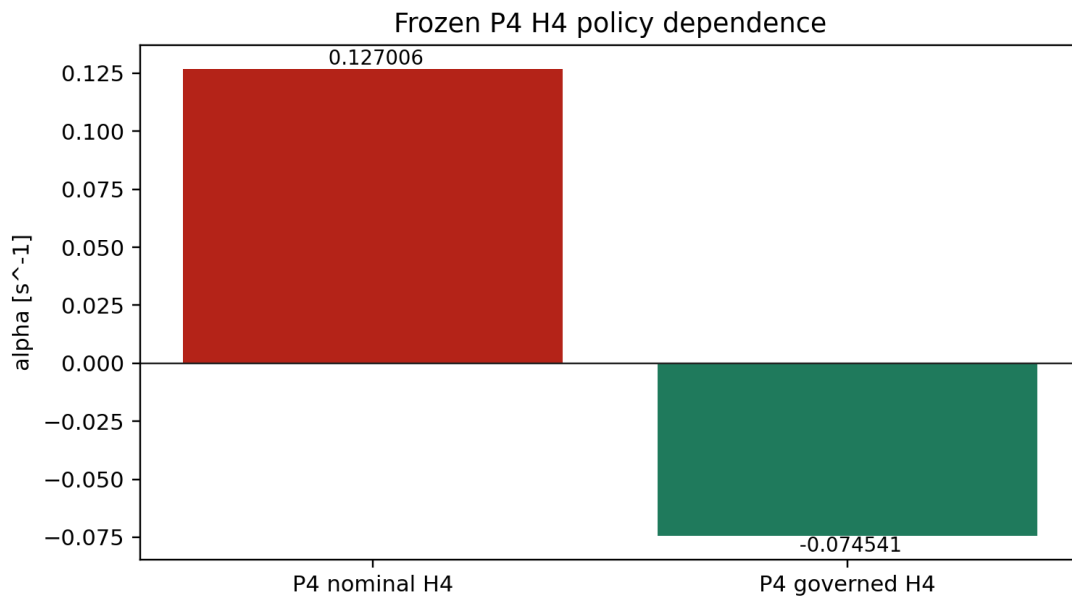


Figure 1. Frozen P4 H4 policy dependence. The governor result is policy-conditioned.

2. Negative and stopped results

- V9: 395/512 correct, 116 false-safe, and 1 false-unstable. The plan's 511/512 and zero-error expectation is REFUTED.
- The exact V9 blocker antichain is not supported.
- PowerDynamics does not reproduce the frozen GFL mechanism because the available tutorial is a different synchronous-machine/governor model.
- No second converter model, new blind topology/model holdout, physical robust radius, common uncertainty envelope, new Julia TDS, EMT, current-limit, DC-link, or hardware certification is claimed.

3. Safe claim

In the frozen IEEE-39 benchmark and stated policy, the spectral/collective closure audit reproduces a P4 minimal incompatibility and the declared frozen nonlinear verdicts. The historical V9 transfer expectation is refuted, the documented governor changes the P4 H4 sign, and robustness, second-model, same-mechanism PowerDynamics, and new blind-holdout claims remain untested or stopped by scope.

4. Reproducibility

Gate 0	reports/GATE0_REPORT.md
Theory tests	logs/theory_tests.log; 4 modules passed
PowerDynamics	raw/powerdynamics/pd39_equilibrium_gate.md
Claim ledger	docs/CLAIM_LEDGER.md
Dashboard	reports/GATE_DASHBOARD.md and .csv
Base	IAS2026_FINAL_SCIENTIFIC_EVIDENCE_FREEZE at b9f274e...